SAI KIRAN GOUD LINGALA

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EDUCATION

University of Dayton - Dayton, OH | **M.S. Mechanical Engineering** | Dec 2025
Concentration in heat transfer, computational fluid dynamics and compressible flow

MLR Institute of Technology - Hyderabad, India | **B.Tech. Aeronautical Engineering** | Jul 2022

TECHNICAL SKILLS

CFD and Thermal: ANSYS Fluent, Simcenter STAR-CCM+, OpenFOAM, ParaView; conjugate heat transfer, thermal resistance networks, liquid cold plate and coolant loop analysis, pressure drop and flow distribution, RANS turbulence modeling (k-omega SST, Realizable k-epsilon, Spalart-Allmaras), moving reference frame and sliding-mesh rotating machinery, transonic and shock-boundary-layer flow, steady and transient internal and external flow

Verification and Validation: Grid Convergence Index, Richardson extrapolation, mesh independence, y+ control and boundary-layer meshing, multi-region interfaces, correlation against published experimental and DNS reference data, signed-deviation reporting, model-form uncertainty quantification, root cause investigation of model discrepancy

Programming and CAD: Python (NumPy, Pandas, Matplotlib, PyTorch), MATLAB, solver journal scripting, physics-informed neural networks (2D Navier-Stokes solver, Re 10), automatic differentiation, Git, Linux; Siemens NX, SolidWorks, CATIA V5/V6, Ansys Discovery / SpaceClaim

SELECTED PROJECTS

NASA Rotor 37 - Transonic Compressor Rotor CFD Validation | Simcenter STAR-CCM+ | github.com/SaiKiran4099/nasa-rotor37-cfd

- Validated a steady RANS model of the NASA Rotor 37 transonic axial compressor against published NASA measurements to within **1.1 percent** on total pressure ratio and **2.2 points** on adiabatic efficiency, building the single-passage rotating domain at **y+ near 1** with **36** cells across the **0.357 mm** tip gap.
- Established turbulence closure rather than mesh as the dominant uncertainty: Spalart-Allmaras against k-omega SST on an identical **3.71 million**-cell mesh gave a **3.9 percent** spread in pressure ratio, against **0.9 percent** from a four-mesh Grid Convergence study, with the closures bracketing the measured value.
- Reported what the model could not reproduce alongside what it could, including a **7 to 11 point** efficiency deficit at **97 percent** span and excess hub total pressure from an unmodeled leakage path.

Liquid-Cooled Cold Plate and Manifold - Thermal Design and CFD Validation | Siemens NX, Simcenter STAR-CCM+ | *Ongoing*

- Derived a design requirement of **0.0938 K/W or less** against a current **0.1032 K/W** conjugate heat transfer result by building the case-to-coolant resistance network from published package and interface data, establishing the cold plate as **94 percent** of the path, and confirmed the design thermally rather than hydraulically limited at **0.0015 percent** pumping power.

Rotating Machinery - CFD Root Cause Diagnosis and Benchmark Validation | Simcenter STAR-CCM+

- Reduced a **2.1x** benchmark over-prediction to within **3 percent** by refuting rotor-stator unsteadiness and turbulence-model bias before isolating a **25 percent** oversized swept volume through dimensional audit, predicting the corrected result analytically before rebuilding and closing steady moving-reference-frame against transient sliding-mesh to **0.74 percent**.

Turbine Blade Conjugate Heat Transfer - Cooling Configuration Trade-Off | ANSYS Fluent, SpaceClaim

- Compared cooling configurations on a high-temperature turbine blade by coupling solid and fluid domains so conduction and convection resolved together rather than assuming an interface heat transfer coefficient, modeling film cooling, internal channels and thermal barrier coatings.

PROFESSIONAL EXPERIENCE

Cyient - Hyderabad, India | **CFD and Aerodynamics Engineer** *(aerodynamic analysis and design)* | Jan 2022 - Jul 2023

- Improved lift coefficient by **7 percent** and lift-to-drag ratio by approximately **11 percent** by diagnosing trailing-edge flow separation to root cause from velocity-field post-processing and implementing a targeted leading-edge geometry modification.
- Achieved grid-convergence agreement within **0.8 percent** across a **14**-case parametric study by applying Grid Convergence Index and Richardson extrapolation over **3** systematically refined meshes.
- Reduced new-case turnaround by approximately **35 percent** by automating boundary-condition setup, multi-case data extraction and report preparation in Python and solver journal files.

University of Dayton - Dayton, OH | **Graduate Teaching Assistant, Heat Transfer** | Aug 2023 - Aug 2025

- Mentored **20+** students through conduction, convection, radiation and internal flow analysis across **2** academic years.